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APPROVAL REPORT

Project No: PR462212

Class: 1630

Product Type: Steel Pipe for Automatic Fire Sprinkler Systems

Product Name: Steel Pipe Manufactured to
ASTM A795 Schedule 10
ASTM A53 & ASTM A795 Schedule 40
Specifications

Name of Report Holder: Tianjin City Baolai Industrial and Trade Co Ltd

Address of Report Holder: NO.6, Haihe Road, Daquizhuang Town,
Jinghai District, Tianjin,
China

Customer ID: 264925

Prepared by

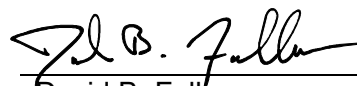


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20 February 2024

Date of Approval

1 INTRODUCTION

- 1.1 Tianjin City Baolai Industrial and Trade Co. requested an FM Approval examination of their products with rated working pressures in accordance with the following:

Model	Product Type	End Connection	Sizes, NPS	Rated Working Pressure	
				psi	kPa
Schedule 10	ASTM A795	Plain End	1, 1-1/4, 1-1/2, 2	175	1205
		Welded	1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6, 8, 10	175	1205
		Rolled Groove			
Schedule 40	ASTM A53	Plain End	1, 1-1/4, 1-1/2, 2	175	1205
		Welded	1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6, 8, 10	200	1375
		Threaded			
		Rolled Groove			
		Cut Groove			
	ASTM A795	Plain End	1, 1-1/4, 1-1/2, 2	175	1205
		Welded	1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6, 8, 10	200	1375
		Threaded			
		Rolled Groove			
		Cut Groove			

- 1.2 These are new products for Tianjin City Baolai Industrial and Trade Co.

- 1.3 This Report may be freely reproduced only in its entirety and without modification.

1.4 Standards

1.4.1 FM Approvals Standards

Title	Number	Issue Date
Steel Pipe for Automatic Fire Sprinkler Systems	1630	June 2021

1.5 Listing

The product(s) will be updated in the Approval Guide, an on-line resource of FM Approvals, as detailed in an attachment at the end of this report. Deletions from any current product listing are shown with strikethroughs and additions to the Approval listing are shown in red text.

- 1.6 In addition to the evaluation of the product itself, a First Audit has been conducted at the manufacturing location(s) for the products discussed in this Report. Please refer to Section 6 of the Report for additional information related to the First Audit.

2 DESCRIPTION

2.1 ASTM A795 / A795M Schedule 10 Sprinkler Pipe

Schedule 10 sprinkler pipe manufactured in accordance with the specifications set forth in American Society of Testing and Materials (ASTM) Standard A795 / A795M has nominal wall thicknesses as detailed below. As part of the evaluation, steel manufactured to the requirements of ASTM A795 / A795M Grades A and B have been considered. The pipe is available in a black, epoxy painted or hot dip galvanized surface finish and the end connections are plain end, beveled (welded) or rolled groove.

ASTM A795 / A795M Schedule 10 Sprinkler Pipe Dimensions		
Nominal Pipe Size inch	Nominal Outside Diameter inch (mm)	Nominal Wall Thickness inch (mm)
1	1.315 (33.4)	0.109 (2.77)
1-1/4	1.660 (42.2)	0.109 (2.77)
1-1/2	1.900 (48.3)	0.109 (2.77)
2	2.375 (60.3)	0.109 (2.77)
2-1/2	2.875 (73.0)	0.120 (3.05)
3	3.500 (88.9)	0.120 (3.05)
3-1/2	4.000 (101.6)	0.120 (3.05)
4	4.500 (114.3)	0.120 (3.05)
5	5.563 (141.3)	0.134 (3.40)
6	6.625 (168.3)	0.134 (3.40)
8	8.625 (219.1)	0.188 (4.78)
10	10.750 (273.0)	0.188 (4.78)

2.2 ASTM A53 / A53M Schedule 40 Sprinkler Pipe

Schedule 40 sprinkler pipe manufactured in accordance with the specifications set forth in American Society of Testing and Materials (ASTM) Standard A53 / A53M has nominal wall thicknesses as detailed below. As part of the evaluation, steel manufactured to the requirements of ASTM A53 / A53M Grades A and B have been considered. The pipe is available in a black, epoxy painted or hot dip galvanized surface finish and the end connections are plain end, beveled (welded), threaded or rolled or cut groove.

ASTM A53 / A53M Schedule 40 Sprinkler Pipe Dimensions		
Nominal Pipe Size inch	Nominal Outside Diameter inch (mm)	Nominal Wall Thickness inch (mm)
1	1.315 (33.4)	0.133 (3.38)
1-1/4	1.660 (42.2)	0.140 (3.56)
1-1/2	1.900 (48.3)	0.145 (3.68)
2	2.375 (60.3)	0.154 (3.91)
2-1/2	2.875 (73.0)	0.203 (5.16)
3	3.500 (88.9)	0.216 (5.49)
3-1/2	4.000 (101.6)	0.226 (5.74)
4	4.500 (114.3)	0.237 (6.02)
5	5.563 (141.3)	0.258 (6.55)

ASTM A53 / A53M Schedule 40 Sprinkler Pipe Dimensions		
Nominal Pipe Size inch	Nominal Outside Diameter inch (mm)	Nominal Wall Thickness inch (mm)
6	6.625 (168.3)	0.280 (7.11)
8	8.625 (219.1)	0.322 (8.18)
10	10.750 (273.0)	0.365 (9.27)
12	12.750 (323.8)	0.406 (10.31)

2.3 ASTM A795 / A795M Schedule 40 Sprinkler Pipe

Schedule 40 sprinkler pipe manufactured in accordance with the specifications set forth in American Society of Testing and Materials (ASTM) Standard A795 / A795M has nominal wall thicknesses as detailed below. As part of the evaluation, steel manufactured to the requirements of ASTM A795 / A795M Grades A and B have been considered. The pipe is available in a black, epoxy painted or hot dip galvanized surface finish and the end connections are plain end, beveled (welded), threaded or rolled or cut groove.

ASTM A795 / A795M Schedule 40 Sprinkler Pipe Dimensions		
Nominal Pipe Size inch	Nominal Outside Diameter inch (mm)	Nominal Wall Thickness inch (mm)
1	1.315 (33.4)	0.133 (3.38)
1-1/4	1.660 (42.2)	0.140 (3.56)
1-1/2	1.900 (48.3)	0.145 (3.68)
2	2.375 (60.3)	0.154 (3.91)
2-1/2	2.875 (73.0)	0.203 (5.16)
3	3.500 (88.9)	0.216 (5.49)
3-1/2	4.000 (101.6)	0.226 (5.74)
4	4.500 (114.3)	0.237 (6.02)
5	5.563 (141.3)	0.258 (6.55)
6	6.625 (168.3)	0.280 (7.11)
8	8.625 (219.1)	0.322 (8.18)
10	10.750 (273.0)	0.365 (9.27)
12	12.750 (323.8)	0.406 (10.31)

3 EXAMINATIONS AND TESTS

3.1 Samples were submitted for examination and testing. The samples were considered to be representative of the product line and were examined, tested, and compared to the manufacturer's drawings. All data remains on file at FM Approvals along with other documents and correspondence applicable to this program.

3.2 All testing and analysis considered appropriate was conducted and verified to be in compliance with the standards defined in Section 1.4.

3.3 Detailed analysis of the examination and testing can be found as an attachment at the end of this Report.

4 MARKING

4.1 Each length of pipe shall be permanently and continuously marked with the following minimum information:

- Manufacturer's name or trademark
- Nominal pipe size and length
- Schedule or model reference
- Rated working pressure
- National or International standard (including grade reference) to which the pipe was manufactured
- Manufacturing Source code
- Production test reference
- Heat number or Run number
- The FM Approvals Certification Mark

4.2 Markings shall be painted, inked or laser printed on the outside surface of each length of pipe at regular intervals. The spacing between the end of one complete marking and the start of the next shall not exceed 12 inches (300 mm).

4.3 Each marking described above shall be legible and durable and shall be applied in any of, or any combination of, the above methods.

5 REMARKS

5.1 The FM Global Property Loss Prevention Data Sheets should be strictly adhered to when installing this product.

5.2 Installations shall comply with the latest edition of the manufacturer's instruction manual.

5.3 Tampering and/or replacement with non-factory components may adversely affect the safe use of the product.

6 SURVEILLANCE AUDIT

The design and manufacturing facilities at the following location(s) are subject to follow-up audit inspections. The facilities and quality control procedures in place have been found to be satisfactory to manufacture product identical to that examined and tested as described in this Report. A revision request form shall be submitted to FM Approvals for requesting any additional manufacturing facilities which are not listed below. The Products discussed in this Report are FM Approved only when designed and manufactured in the following facilities:

Design & Manufacturing

Tianjin City Baolai Industrial and Trade Co Ltd
NO.6, Haihe Road, Daqiu Zhuang Town
Jinghai District, Tianjin City
China

Grooving, Threading & Galvanizing

Tianjin Baolaili Galvanized Steel Pipe Co
500m Northeast of Gangjingxian Line & Youfa Rd
Taiping Village, Daqiu Zhuang Town
Jinghai District, Tianjin City
China

Epoxy Coating

Tianjin Xinjinyu Anticorrosion & Insulation Steel Tube Co
No.10, Haihe Road, Daqiu Zhuang Town
Jinghai County, Tianjin City
China

7 MANUFACTURER'S RESPONSIBILITIES

- 7.1** Documentation considered critical to this Approval is on file at FM Approvals and is listed in the Documentation File, Section 8, of this Report. No changes of any nature shall be made unless notice of the proposed change has been given and written authorization obtained from FM Approvals. The revision request form shall be forwarded to FM Approvals as notice of proposed changes.
- 7.2** The manufacturer is responsible for control of the product marking and installation instructions for the product.
- 7.3** The manufacturer shall provide installation, operating, and maintenance manual(s) with each system.
- 7.4** The manufacturer is responsible for performing the Manufacturing and Production Tests specified in the Standard defined in Section 1.4 of this Approval Report.

8 DOCUMENTATION FILE

All pertinent Report documents are outlined in the ATTACHMENTS list below.

9 CONCLUSION

The Products described in Section 1 of this Report meet FM Approvals requirements when manufactured at the facilities detailed in Section 6 of this Report. Since a duly signed Master Agreement is on file for this manufacturer, Approval is effective the date of this report.

PROJECT DATA RECORD: PR462212

ATTACHMENTS:	Appendix A	Approval Guide Listings
	Appendix B	Detailed Analysis
	Appendix C	Critical Document List (CDL)

APPROVAL GUIDE LISTING**SCHEDULE 10**

Fire Protection – Automatic Sprinkler Systems – Pipes and Fittings for Aboveground – Steel Pipe – Schedule 10 – Normal connections are welded, rolled groove or plain end

Tianjin City Baolai Industrial and Trade Co Ltd

NO.6, Haihe Road, Daqiu Zhuang Town, Jinghai District, Tianjin City, China

Product	Nominal Pipe Size, in.	Rated Working Pressure, psi	Rated Working Pressure, kPa	Listing Country	Certification Type	Class of Work
Schedule 10 ^{a, c, d, e, g, m, n, s}	1, 1-1/4, 1-1/2, 2	175	1205	China	FM Approved	1630-Steel Pipe Spklr Sys Serv
Schedule 10 ^{a, c, d, e, m, n, s, t}	1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6, 8, 10	175	1205	China	FM Approved	1630-Steel Pipe Spklr Sys Serv

Notes:

- a - FM Approved for use with FM Approved pipe couplings on rolled grooves.
- c - FM Approved for use in welded systems when supplied with standard bevel on ends.
- d - When hot dip galvanized by factory, the sprinkler pipe is FM Approved for dry pipe systems.
- e - Any FM Approved pipe coupling or fitting suitable for use with Schedule 10 pipe may be used with this product at the lower rated pressure of the pipe or the fitting.
- g - FM Approved for use with plain-end fittings.
- m - FM Approved for use in all steel sprinkler systems composed of uncoated steel pipe
- n - FM Approved for use in hybrid sprinkler systems composed of uncoated steel and plastic piping.
- s - Manufactured out of ASTM A795 / A795M Grade A or B steel.
- t - Pipe 8 in. or larger is FM Approved with a wall thickness of 0.188 in., which is above ASME B36.10 specifications for Schedule 10 pipe.

APPROVAL GUIDE LISTING**SCHEDULE 40**

Fire Protection – Automatic Sprinkler Systems – Pipes and Fittings for Aboveground – Steel Pipe – Schedule 40 – Normal connections are threaded, welded, rolled or cut groove or plain end

Tianjin City Baolai Industrial and Trade Co Ltd

NO.6, Haihe Road, Daqiu Zhuang Town, Jinghai District, Tianjin City, China

Product	Nominal Pipe Size, in.	Rated Working Pressure, psi	Rated Working Pressure, kPa	Listing Country	Certification Type	Class of Work
Schedule 40 ^{a, b, c, d, e, g, m, n, p, s}	1, 1-1/4, 1-1/2, 2	175	1205	China	FM Approved	1630-Steel Pipe Spklr Sys Serv
Schedule 40 ^{a, b, c, d, e, m, n, p, s}	1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6, 8, 10	200	1375	China	FM Approved	1630-Steel Pipe Spklr Sys Serv

Notes:

- a - FM Approved for use with FM Approved pipe couplings on rolled grooves.
- b - FM Approved for use with FM Approved pipe fittings when threaded
- c - FM Approved for use in welded systems when supplied with standard bevel on ends.
- d - When hot dip galvanized by factory, the sprinkler pipe is FM Approved for dry pipe systems.
- e - Any FM Approved pipe coupling or fitting suitable for use with Schedule 10 pipe may be used with this product at the lower rated pressure of the pipe or the fitting.
- g - FM Approved for use with plain-end fittings.
- m - FM Approved for use in all steel sprinkler systems composed of uncoated steel pipe
- n - FM Approved for use in hybrid sprinkler systems composed of uncoated steel and plastic piping.
- p - Manufactured out of ASTM A53 / A53M Grade A or B steel.
- s - Manufactured out of ASTM A795 / A795M Grade A or B steel.

DETAILED ANALYSIS**1 EXAMINATION**

The manufacturer provided samples of their pipe as detailed below for examination and testing. The samples were considered to be representative of the product line and were examined, tested, and compared to the manufacturer's drawings. All data is on file at FM Approvals along with other documents and correspondence applicable to this program.

FM 1630 Sample Requirements – Steel Pipe			
Test	Pipe Class / Model	NPS, inches	Fitting Type / Model
Dimensional Examination	ASTM A53 Sch 10	2	-
		4	-
		6	-
		8	-
		10	-
	ASTM A795 Sch 40	2	-
		4	-
		6	-
		8	-
		10	-
Hydrostatic Strength / Bending Moment	ASTM A53 Sch 10	2	Flexible – Meide 1N
			Flexible – Victaulic 75
			Rigid – Meide 1G
			Rigid – Victaulic 005H
		4	Flexible – Meide 1N
			Flexible – Victaulic 75
			Rigid – Meide 1G
			Rigid – Victaulic 005H
		6	Flexible – Meide 1N
			Flexible – Victaulic 75
			Rigid – Meide 1G
			Rigid – Victaulic 005H
		8	Flexible – Meide 1N
			Flexible – Victaulic 75
			Rigid – Meide 1G
			Rigid – Victaulic 005H
		10	Flexible – Meide 1N
			Flexible – Victaulic 77
			Rigid – Meide 1G
			Rigid – Victaulic 07

FM 1630 Sample Requirements – Steel Pipe			
Test	Pipe Class / Model	NPS, inches	Fitting Type / Model
Hydrostatic Strength / Bending Moment	ASTM A795 Sch 40	2	Threaded – Meide 220
			Threaded – Langfang CPL20
			Flexible – Meide 1N
			Flexible – Victaulic 75
			Rigid – Meide 1G
			Rigid – Victaulic 005H
		4	Threaded – Meide 220
			Threaded – Langfang CPL20
		6	Flexible – Meide 1N
			Flexible – Victaulic 75
			Rigid – Meide 1G
			Rigid – Victaulic 005H
		10	Flexible – Meide 1N
			Flexible – Victaulic 77
			Rigid – Meide 1G
			Rigid – Victaulic 07
Rot. Bending Moment	ASTM A53 Sch 10	4 x 3	Mech. Tee – Meide 3G
			Mech. Tee – Victaulic 920
Vibration Resistance	ASTM A53 Sch 10	2	Flexible – Meide 1N
			Flexible – Victaulic 75
	ASTM A795 Sch 40	2	Flexible – Meide 1N
			Flexible – Victaulic 75
			Threaded – Meide 220
			Threaded – Langfang CPL20
Marking Durability	ASTM A53 Sch 10	4	Black Finish
			Epoxy Finish
			Galvanized Finish
Corrosion Resistant Coatings	ASTM A53 Sch 10	4	Galvanized Finish
	ASTM A795 Sch 40	4	Galvanized Finish

2 DESCRIPTION

2.1 Hydrostatic Strength

Samples as detailed above were filled with water, cleared of all entrapped air, and subjected to four times their rated working pressure for 5 minutes. There were no signs of cracking, rupture or permanent distortion of the assembly as a result of this test. These results are considered satisfactory.

2.2 Bending Moment Resistance

Samples as detailed above were filled with water, cleared of all entrapped air, and subjected to their rated working pressure. The assembly was centered in a fixture that provided support at a spacing of 24 inches (0.6 m) for pipes 6 inch NPS and smaller or a spacing of 48 inches (1.2 m) for pipes 8 inch NPS and larger and was subjected to the required bending moment. The bending moment was calculated based on water filled Schedule 40 steel sprinkler pipe, a maximum hanger spacing of 12 or 15 ft (3.6 or 4.6 m) depending on pipe size, the assumption of a missing hanger and a safety factor of two. There was no observed evidence of leakage, cracking, or rupture as a result of this test. These results are considered satisfactory.

2.3 Rotational Bending Moment Resistance

Samples as detailed above were assembled into a tee-shaped assembly. The assembly was filled with water, cleared of all entrapped air, and subjected to its rated working pressure. With the run pipe securely restrained, the assembly was subjected to a rotational bending moment in two directions; one tending to rotate the fitting around the centerline of the run pipe, and the other coplanar with the axis of the run pipe. The bending moment was calculated based on water filled Schedule 40 steel sprinkler pipe, a maximum hanger spacing of 12 or 15 ft (3.6 or 4.6 m) depending on pipe size, the assumption of a missing hanger and a safety factor of two. There was no observed evidence of leakage, cracking, or rupture as a result of this test. These results are considered satisfactory.

2.4 Vibration Resistance

Samples as detailed above were filled with water, cleared of all entrapped air, and subjected to an internal pressure of 80 psi (550 kPa) which was maintained throughout the vibration test sequence. The samples were then subjected to the following vibration conditions:

Amplitude, inch (mm)	Total Displacement, inch (mm)	Frequency, Hz	Time, hours
0.020 (0.51)	0.040 (1.02)	18 – 37 (Variable)	5
0.035 (0.90)	0.070 (1.78)	18 – 37 (Variable)	5
0.010 (0.25)	0.020 (0.51)	28	5
0.020 (0.51)	0.040 (1.02)	28	5
0.075 (1.91)	0.150 (3.81)	28	5

After the completion of the vibration sequence, the samples were subjected to four times their rated working pressure for 5 minutes. There was no observed evidence of leakage, cracking, or rupture as a result of this test. These results are considered satisfactory.

2.5 Marking Durability

Samples as detailed above were subjected to water immersion for 5 minutes on each of 10 working days. After the 10 day period, the samples were removed from the water and were rubbed vigorously on the marked surface with cotton or paper towels. Subsequently, the print line on the samples were covered in petroleum based grease for a period of 24 hours, after which the grease was removed with cotton or paper towels. There was no evidence of smearing, bleeding, or removal of the markings as a result of this test. These results are considered satisfactory.

2.6 Corrosion Resistant Coatings

Samples as detailed above were subjected to corrosion resistant coating testing. The galvanized coating thickness was measured using a calibrated coating thickness gauge at a minimum of 4 points for 2 inch NPS (DN50) and smaller pipe and at a minimum of 6 points for larger pipe. Measurements were taken around the internal and external surfaces of each sample and the readings were averaged and compared to the requirements. The average coating thickness was 1.8 oz/ft² (0.55 kg/m²) or greater and no individual measurement was less than 1.5 oz/ft² (0.46 kg/m²). These results are considered satisfactory.

3 **CONCLUSION**

Based on the above results, no additional tests were deemed necessary.

CRITICAL DOCUMENT LIST (CDL)

The following drawings describe the steel sprinkler pipe discussed in this Report and are filed under Project Identification number 462212.

Drawing No.	Description	Rev. Level
BL-MS-001	Marking Sample	1
BLL CHX06-01	Groove Dimensions	A/1
BLL CHX06-02	Thread Dimensions	A/0
P-1-001	Plain End Steel Pipe	0
W-1-001	Schedule 40 Steel Pipe	0
W-1-002	Schedule 10 Steel Pipe	1